UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	INTRODUCTION TO INTERACTIVE MULTIMEDIA
COURSE CODE	CSC413
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages.

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PART A (15 MARKS)

State TRUE (T) or FALSE (F) for the following statements.

1. Path animation moves an object along a predefined route.
2. A prototype is an early model of a product.
3. 3D animation does not use modelling.
4. MPEG is a common family of video compression formats.
5. A multimedia project always uses every media element.
6. A URL identifies the address of a web resource.
7. Chroma key compositing replaces a colour background with another image.
8. HTML is used to structure content on web pages.
9. Interactivity level is irrelevant when choosing an authoring tool.
10. Streaming requires the whole file to download before playback.
11. Higher resolution always means a smaller file.
12. A vector image becomes pixelated when enlarged.
13. Frame rate is the number of video frames displayed per second.
14. Keyframes define important positions in an animation.
15. Sampling converts a continuous sound wave into digital values.

PART B (15 MARKS)

Select the most appropriate answer from Table A for each of the following questions.

Table A. Matching options

A. Vector	B. Cel animation	C. Graphics tablet
D. Target audience	E. Serif	F. Multimedia
G. Keyframe	H. Scanner	I. Tweening
J. Bitmap	K. Sampling	L. Plotter
M. RGB	N. Frame rate	O. Storyboard

1. image made of pixels
2. frames displayed per second
3. visual sequence plan
4. image made of mathematical objects
5. font with small strokes at character ends
6. important frame in an animation
7. generating in-between frames
8. input device that digitises documents
9. additive colour model used by screens
10. hand-drawn frame-by-frame animation
11. converting sound into digital values
12. output device for large drawings
13. integration of text, graphics, audio, video and animation
14. intended users of a product
15. input device for drawing

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PART C (30 MARKS)

Choose the correct answer for each question.

1. A presentation where the user controls the sequence is called:
   A. Linear
   B. Static
   C. Printed
   D. Non-linear

2. Hypermedia is best described as:
   A. A single video file
   B. A printed manual
   C. Text-only media
   D. Multimedia with hyperlinks

3. Which is an example of linear multimedia?
   A. A game with choices
   B. A cinema presentation
   C. An interactive kiosk
   D. A website with menus

4. Interactivity in multimedia means:
   A. The system plays automatically
   B. Only text is shown
   C. The user can control content and flow
   D. There is no navigation

5. Which format is commonly used for bitmap images?
   A. MPEG
   B. PNG
   C. HTML
   D. MIDI

6. RAM is:
   A. A web browser
   B. Permanent storage
   C. Temporary working memory
   D. An input device

7. An authoring tool is used to:
   A. Only browse the web
   B. Only record audio
   C. Only print text
   D. Assemble multimedia elements

8. Which is a benefit of multimedia in education?
   A. It uses only text
   B. It replaces all teachers
   C. It removes all interaction
   D. It combines visuals and sound

9. Sampling converts:
   A. Digital video to text
   B. Text to speech
   C. Images to vectors
   D. Analog sound to digital values

10. Which best defines multimedia?
    A. A single image
    B. A printed book
    C. Text, graphics, audio, video and animation
    D. Text only

11. Convergence in multimedia refers to:
    A. Printing images
    B. Merging of media and computing
    C. Deleting media files
    D. Recording audio only

12. Morphing changes:
    A. One shape into another
    B. Font size
    C. File format
    D. Audio pitch

13. A hyperlink links to:
    A. Only audio
    B. Only images
    C. Only text
    D. Another resource

14. Which is NOT a multimedia element?
A. Text
B. Audio
C. Compiler
D. Animation

15. Copyright protects:
A. Computer hardware
B. Internet speed
C. File names
D. Original creative works

16. A graphics tablet is used for:
A. Printing
B. Sound output
C. Drawing input
D. Web browsing

17. Which is an output device?
A. Mouse
B. Plotter
C. Keyboard
D. Scanner

18. Which is an authoring tool category?
A. CMYK
B. RGB
C. Card-based
D. JPEG

19. HTML is used to:
A. Store audio samples
B. Model 3D objects
C. Structure web content
D. Compress video

20. Anti-aliasing is used to:
A. Increase file size only
B. Compress video
C. Smooth jagged edges
D. Add sound

21. Which image type scales without losing quality?
A. Bitmap
B. Vector
C. Photograph
D. Scan

22. MIDI stores:
A. Video frames
B. Images
C. Musical notes and instructions
D. Digitized audio samples

23. A higher sampling rate generally:
A. Changes colour
B. Reduces accuracy
C. Removes audio
D. Improves accuracy

24. Augmented reality:
A. Replaces the real world entirely
B. Only uses audio
C. Adds digital information to the real world
D. Only uses text

25. A storyboard is:
A. A video codec
B. A visual sequence plan
C. A colour model
D. A sound file

26. A browser is used to:
A. Display web pages
B. Scan images
C. Print documents only
D. Edit audio

27. A serif font has:
A. No strokes
B. Small strokes at character ends
C. Only uppercase letters
D. No spacing

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28. Hertz measures:

A. Resolution
B. Frequency
C. File size
D. Loudness

29. Hypertext contains:

A. No text
B. Links to other content
C. Only audio
D. Only images

30. Card-based authoring tools are suited to:

A. 3D modelling only
B. Messages with a beginning and end
C. Audio editing only
D. Live video only

PART D (40 MARKS)

QUESTION 1

Text is an essential element in a multimedia project. Its shape, size, design, colour, arrangement and linking all affect the message.

a) State TWO reasons sans serif is mostly used for screen display and serif is used for printed material.

(3 marks)

b) Describe TWO roles of hypertext in multimedia.

(3 marks)

c) Explain how text can have a massive impact on an informative multimedia project in the link and HTML context.

(4 marks)

QUESTION 2

A shopping complex requires a kiosk that gives customers location and retailer information.

a) State TWO important project team roles for completing the kiosk.

(3 marks)

b) Design a low-fidelity prototype for a pre-school courseware home page. The page must show topic menus, the course title and the school logo.

(3 marks)

c) Discuss TWO factors that lead to the wide use of mobile devices in games and shopping applications.

(4 marks)

QUESTION 3

A company wants an interactive product demonstration that runs on a kiosk and on the web.

a) State TWO features to consider when choosing authoring software.

(2 marks)

b) Explain the interactivity level suitable for the demonstration.

(4 marks)

c) Explain TWO authoring tool categories appropriate for this project.

(4 marks)

QUESTION 4

You are required to animate a company logo. The logo depicts a planet orbiting the sun.

a) State TWO animation file formats suitable for this project.

(2 marks)

b) Describe the steps of cel animation and ONE advantage of using it.

(3 marks)

c) Differentiate 2-D and 2.5-D animation with ONE example for each.

(5 marks)

END OF QUESTION PAPER

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